

ANDREA AITA

Data & Information Strategy | Digital Delivery Lead

London, United Kingdom | +44 7542 546566 | andrea.aita@libero.it

[linkedin.com/in/andrea-aita-0b0794123](https://www.linkedin.com/in/andrea-aita-0b0794123)

BEP Generator - ISO 19650 Compliant BIM Execution Plans | AI-Powered

PROFESSIONAL SUMMARY

Results-driven digital transformation leader with 10 years delivering high-impact infrastructure programmes across energy, nuclear, and construction sectors. Currently seconded to National Grid's £9bn Great Grid Partnership programme, leading digital strategy implementation across 14 transmission projects. Proven track record: achieved 50% reduction in design and project delivery errors, trained 500+ professionals at Heathrow and delivered 15% operational efficiency improvements in nuclear projects. Expert in data governance, ISO 19650, Common Data Environments, and asset information strategy from construction through to operations; with hands-on experience in regulated nuclear environments requiring the highest standards of information assurance.

PROFESSIONAL EXPERIENCE

► Associate Director – Digital Lead (Energy-Nuclear Division)

AECOM | Dallas, Texas (USA) - UK Operations - London | American multinational engineering and infrastructure company ~51,000 employees | Revenue: \$14-15 billion USD (2024)

August 2024 – Present

Leading digital strategy and data governance for AECOM's energy-nuclear division, including digital transformation operations for Sellafield (UK public sector client). Defining information management standards, coordinating advanced technologies, and driving operational efficiency improvements across critical national infrastructure assets.

Key achievements:

- Led technical teams delivering digital services across the energy-nuclear portfolio, increasing operational efficiency by up to 15% (verified by internal project metrics)
- Established data governance frameworks and ISO 19650 information management standards for highly regulated nuclear projects, ensuring compliance with UK nuclear security and information assurance requirements
- Provided strategic advisory on digital tool adoption, process automation, and information lifecycle management for long-duration critical assets

BIM & Digital Delivery Lead (Consultancy Secondment to National Grid)

National Grid - Great Grid Partnership Programme | London, UK | British multinational leader in energy transmission/distribution ~31,600 employees | Revenue: £18+ billion (2024)

September 2024 – Present (seconded from AECOM)

Representing National Grid as client-side digital lead, providing strategic guidance on digital engineering methodologies to optimise asset development and lifecycle management across the Great Grid Partnership (GGP) infrastructure programme. Conducting digital maturity assessments, developing transformation roadmaps, and implementing Common Data Environment (CDE) solutions across multi-project portfolios.

Key achievements:

- Led the development of the Common Data Environment solution for the Great Grid Partnership framework, coordinating digital collaboration across strategic delivery partners including AECOM, WSP, Hitachi, Laing O'Rourke, and Vinci Construction
- Implemented ISO 19650 frameworks across 14 infrastructure projects within the onshore GGP portfolio (electrical transmission, substations, renewables)

- Developed digital strategies including the NG Object Type Library, CDE technical integrations, and paperless construction site operations, delivering scalable solutions such as 20% reduction in downtime through data-driven analysis of design deliverables
- Collaborated with product management, information management, GIS, and solution architecture leads to ensure end-to-end digital integration

► Senior Digital Consultant – Real Estate Digital EMEA

Turner & Townsend | London, UK | Global programme & cost management consultancy | ~22,000 employees | Revenue: \$1.9bn (2023) | 60% acquired by CBRE in 2021

January 2022 – July 2024

Senior consultant responsible for BIM strategy implementation and digital transformation on large-scale, complex infrastructure and real estate projects across EMEA. Developed and deployed ISO 19650-compliant frameworks, managed Common Data Environments (CDE), and provided strategic advisory on digital tool adoption, real-time data analytics and process automation.

Digital Delivery Lead (Consultancy Secondment to Heathrow Airport)

Heathrow Airport Ltd. | London, UK | World's busiest airport by international passengers | ~76,000 people on campus | Revenue >£3bn annual revenue | £11.3bn capital investment programme (2020–2025)

September 2022 – July 2024 (seconded from Turner & Townsend)

Appointed Digital Delivery Lead for the client, overseeing digital information management across six concurrent construction programmes within the H7 investment period. Defined, developed and governed the ISO 19650-compliant Common Data Environment (Autodesk Construction Cloud), coordinated multi-disciplinary stakeholders and enforced robust information governance on critical national infrastructure projects.

Key achievements:

- Achieved a 12% productivity increase through redesigned processes, digital workflow optimisation and automation
- Designed and delivered training on ISO 19650 standards and information management best practices to over 500 professionals
- Reduced design errors at delivery stage by up to 50% through rigorous ISO 19650 Quality Assurance processes (independently verified by internal audits)
- Led the successful transition of six programmes to a new Common Data Environment, significantly improving collaboration, version control and governance

Other projects at Turner & Townsend:

- Google Real Estate & Data Centres EMEA: Authored the complete set of ISO 19650 information requirements (EIR, AIR, OIR, PIR) for Google's global real estate portfolio, including the KGX1 campus and multiple EMEA hyperscale data centres
- UK Environment Agency – “Consensus” Dashboard (Q1–Q2 2022): Designed and delivered a data analysis & report tool using Power BI dashboards analysing client satisfaction surveys, with thematic insights on innovation and sustainability
- Blackstone: Provided project information management leadership for the development of their new European headquarters in London
- Grosvenor: Delivered full BIM coordination and information management services for a high-profile central London property development

► BIM & Information Manager

HA Associates | London, UK | Boutique consultancy specialising in BIM/digital transformation for AEC and luxury hospitality | Focus: innovative and sustainable design

March 2021 – January 2022

Led the firm-wide implementation and governance of BIM processes, developed computational design and automation solutions (Dynamo/Python), and created comprehensive training programmes to upskill the technical team.

Key Projects and achievements:

- Red Sea Development & Amaala Masterplan (Saudi Arabia): Delivery-partner BIM and information management lead; authored project BIM standards, trained multidisciplinary teams, and automated CDE workflows for multi-billion-dollar luxury hospitality mega-projects

- Luxury hospitality portfolio: Ritz-Carlton Diriyah (Riyadh), Avani Palm View (Dubai), multiple Abu Dhabi developments
- Completed full strategic BIM transformation in 9 months: grew BIM department from 5 to 20 staff and delivered 20% operational efficiency improvement

► BIM & Design Systems Coordinator

Foster + Partners | London, UK | World-leading international architecture practice founded by Lord Norman Foster (AJ100 #1) | ~1,700 employees | Revenue: £653m

October 2018 – March 2021

Coordinated BIM delivery on iconic global architectural projects, authored advanced automation scripts (Dynamo/Python) to streamline multidisciplinary workflows, and ensured on-time, on-budget project delivery.

Key Projects:

- Marseille Provence Airport: BIM coordination for 20,000 m² terminal extension and refurbishment; clash detection scripting reduced analysis time by 25%
- China Merchants Bank Global HQ, Shenzhen: Full BIM coordination across 400m+ high-rise office tower, residential and retail podium in Shenzhen Bay
- Bloomberg European HQ (interiors), Amaravati Government Complex (India), The Tulip (London), Battersea Power Station Phase 3

► Civil Engineer – Structural Design & BIM Manager

Mercitalia Services & Technologies (Ferrovie dello Stato Italiane Group) | Italy | Railway logistics and industrial maintenance | ~1,000 employees (S&T division) | Group: ~82,000 employees, €16.5bn revenue

March 2017 – September 2018

Structural engineer specialising in reinforced concrete FEM analysis and LOD 400 BIM modelling for railway maintenance facilities and industrial plants (OML centres, DDM Mestre-Venice, MCPTC Verona, OMV Mestre). Provided on-site technical support.

► Junior Civil Engineer – Site Assistant & BIM Coordinator

I.CO.P S.p.A. | Italy & Denmark | Specialist infrastructure and marine contractor | ~1,100 employees | Revenue: €343m

July 2016 – March 2017

Site assistant and BIM coordinator on Cityringen Metro extension (M3/M4 lines, 17 stations), Copenhagen. Managed As-Built modelling, QA/QC for offshore piling and Cutter Soil Mixing operations at Kongens Nytorv and Gammel Strand stations. Additional site planning and logistics for new Trieste port platform.

EDUCATION

Master's Degree in Building Engineering & Architecture

5-year integrated programme - Università degli Studi di Padova | Italy

Grade: 106/110

Chartered Civil Engineer (Ingegnere Civile - Sezione A) | Ordine degli Ingegneri della Provincia di Udine - Sezione A, Settore Civile e Ambientale - Italy

CERTIFICATIONS & AWARDS

Sep 2024	IBM Data Science Professional Certificate – Professional certificate in data science, machine learning and Python/R toolkit
Aug 2024	UK BPSS Security Clearance (active) – UK national security clearance for critical infrastructure and nuclear projects
Dec 2023	Lean Six Sigma Green Belt – Certification in process optimisation and variability reduction for complex projects
Nov 2023	Autodesk 40 Under 40 Champions of Construction Award – International recognition for innovators under 40 in the AEC (Architecture, Engineering, Construction) sector
Mar 2023	RICS Certified BIM Professional – ISO 19650 Implementation – Royal Institution of Chartered Surveyors

AI & PRODUCT DEVELOPMENT

BEP Suite - *Full-Stack Platform for ISO 19650 BIM Execution Plans, EIRs and MIDPs (Beta)*

Independently designed and developed a web-based platform that automates ISO 19650-compliant BIM Execution Plan creation using locally-hosted AI — no confidential project data leaves the network. Core features include AI-powered EIR document analysis with structured data extraction, guided BEP authoring across 24+ field types, and automated TIDP-to-MIDP consolidation

Built using agentic AI development tools (Claude Code, GitHub Copilot, Cursor), demonstrating hands-on proficiency with AI-augmented workflows and the ability to deploy local LLMs within confidentiality-sensitive infrastructure environments.

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TECHNICAL SKILLS

Digital Transformation & Strategic Consulting

- Data governance and digital strategy implementation for energy (nuclear, transmission/distribution, renewables) and infrastructure sectors
- Advisory on process optimisation, operational risk reduction and complex multi-project programme management

Information Management & BIM

- ISO 19650 frameworks: authoring EIR, AIR, OIR, PIR, BEP and asset information strategies
- Common Data Environment (CDE) implementation and management: Autodesk Construction Cloud, BIM 360, ProjectWise, SharePoint, Procore

AI-Assisted Development & Agentic AI

- Proficient with AI-augmented development: Claude Code (agentic coding), GitHub Copilot, Cursor IDE
- Local LLM deployment and prompt engineering (Ollama, FastAPI, streaming inference pipelines)
- Application development: React, Node.js/Express, Python, SQLite, Docker, Vite

Automation & Computational Design

- Advanced scripting: Dynamo, Grasshopper, Rhino, Inside, Python (Revit API)
- Workflow automation and elimination of manual errors in complex modelling environments

Data Analytics & Reporting

- Python (pandas, numpy, scikit-learn), Power BI, advanced Excel, Azure Cloud for real-time analytics and KPI dashboards
- Machine learning and data science applications for quantitative strategic decision support

Digital Twin & Asset Lifecycle Management

- Deployment of digital twin technologies for asset monitoring, predictive simulation and full lifecycle management
- Data interoperability and distributed team collaboration on mission-critical infrastructure

Software & Technical Solutions

BIM & Design: Autodesk Revit, AutoCAD, Navisworks, Civil 3D, Rhino 3D, Grasshopper | **CDE:** Autodesk Construction Cloud, BIM 360, ProjectWise, Procore | **Analytics:** Power BI, Tableau, Microsoft Excel, Python (pandas, matplotlib, scikit-learn) | **AI Dev:** Claude Code, GitHub Copilot, Cursor, Ollama, VS Code | **PM:** MS Project, Primavera P6, JIRA

LANGUAGES

Italian | Native

English | Fluent (C2)

German | Elementary (A2)